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NPS/003/012 – Technical Specification for CER Transformers

1. Purpose

This specification details the technical requirements for Continuous Emergency Rated (CER) transformers for use at 66kV or 33kV on the CE Electric UK distribution networks.

This document supersedes the following document, all copies of which shall be destroyed.

Ref	Version	Title
NPS/003/012	1	Technical Specification for CER Transformers
		NEDL Turnkey - Transformer CER Specification

2. Scope

This specification covers the technical requirements for 66kV and 33kV CER transformers for use on the CE Electric UK distribution networks. Also included are the technical specifications for neutral displacement voltage transformers (NVDT) and tank mounted auxiliary transformers, in appendices 5 and 6.

This specification does not cover Continuous Maximum Rated (CMR) transformers.

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3. Technical Requirements

3.1 Overview

CER transformers might be required for the following type of substation:

- Indoor substations
- Outdoor substations where exposed HV conductors are not permitted
- Outdoor situations where exposed HV connections are required

3.2 Technical Specification

3.2.1 Overall

The equipment shall comply fully with Energy Networks Association Technical Specification ENATS 35-2: Emergency Rated System Transformers 33/11.5kV Delta/Star and Star/Star Connected: Issue 5 unless varied by this specification in which case this specification shall take precedence.

The equipment shall comply fully with current versions of all other relevant IEC International Standards, British Standard Specifications or equivalent Euro-Norms, and Energy Networks Association Technical Specifications, except where varied by this standard.

The Equipment must comply with all relevant UK health and safety legislation.

3.2.2 Variations to ENATS 35-2

Although ENATS 35-2 restricts itself to 33/11.5kV transformers, the principles shall also be applied to CER transformers with a primary voltage 66kV, CER transformers with a secondary voltage of 20kV, and transformers with specified CER power rating different to those in ENATS 35-2.

Where appropriate the following variations, additions or clarifications to ENATS 35-2 are referenced to the clause numbers used in issue 5 of that document:

4.1 Rating: The ambient temperature shall be 30°C.

4.1 Rating: Table 1: Standard values of rated power: An additional rating of 15MVA ONAN, 30MVA CER, shall be added.

4.2 Loading Cycle: In addition to a graph of CER rating versus ambient temperature the following graphs shall also be provided to CE Electric UK for each transformer:

- ONAN rating versus ambient temperature
- CMR rating versus ambient temperature.

These graphs shall be verified against temperature rise test results when these tests are undertaken.

5.4 The tapping range shall be one of the two ranges given in ENATS 35-2 as selected in this Appendix 1 of this document.

a) $(\pm 8 \times 1.25\%) \pm 10\%$ in 16 steps

or

b) $(+4 \times 1.43\% / -12 \times 1.43\%) + 5.72 / -17.16\%$ in 16 steps

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5.5 Specification of short-circuit impedance: Impedance shall be as specified in ENATS 35-2. Where an option exists within ENATS 35-2 the required option will be confirmed in Appendix 1 of this document.

5.5 Specification of short-circuit impedance: Figure 1: Impedance envelopes for emergency rated system transformers operating two in parallel:

For 15/30MVA units the impedance envelope criteria are:

		Plus Tap	Mid Tap	Minus Tap
Type A	Max	27	25.5	25
	Min	21	23	23

		Plus Tap	Mid Tap	Minus Tap
Type B	Max	33	32	31
	Min	27	29	29

5.6 Load loss and temperature rise: Losses shall be capitalised using the loss capitalisation values in Appendix 1 of this document:

Lifetime Cost =

Purchase price + ((No load loss kW x No load £/kW) + (Load loss kW x Load loss £/kW)).

Note that the CE Electric UK loss £/kW figures incorporate utilisation factor and timespan.

6.0 Connection and phase displacement symbols for three phase transformers:

The vector groups required for use on CE Electric UK's networks are:

Voltage (kV)	YEDL	NEDL
33/11.5	YNyn0	YNyn6 or Dyn11
66/11.5	YNyn0	Dyn11/Dyn1*
66/20	--	YNyn0

*the transformer shall have vector changeover links

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7.1 Rating Plate: Information to be given in all cases:

In addition the rating plate should state:

The ONAN rating at 30°C ambient temperature
The CER rating at 30°C ambient temperature
The equivalent CMR rating at 30°C ambient temperature
The nominal tap position
The tapping range shall be expressed in the format
+ / - 10% in 16 steps (+ / - 8 x 1.25%) or
+ 5.72 / -17.16% in 16 steps (+ 4 x 1.43% / - 12 x 1.43%)

8.2 Oil preservation system: Where the oil preservation system is a free breathing system the breathing pipe shall be terminated with a 1" (one inch) BSP thread to allow the use of a cartridge system of desiccant filters, such as the Brownell Type W system or equivalent. The pipework shall be arranged to allow these cartridges to be changed easily from ground level. These requirements apply to all oil systems that form part of the transformer installation.

10.1 Tests: General: The mechanical pressure tests shall be completed before any electrical type or routine tests.

10.1 Tests: General: The following shall apply to Table 2

The Rated lightning impulse (LI) for 11kV shall be 95kV (peak) and 38kV (rms)
The Rated lightning impulse (LI) for 20kV shall be 125kV (peak) and 50kV (rms)
The Rated lightning impulse (LI) for 66kV shall be 325kV (peak) and 140kV (rms)

10.1.1 Routine tests: Item (h): Two frequency response analysis (FRA) tests shall be carried out on every CER transformer with an input voltage of 66kV or above. The first FRA test shall be done in the factory after all other routine tests and before the transformer is moved or dismantled. The second test shall be done on-site after offloading and installation/assembly, but before on-site commissioning testing.

10.1.1 Routine tests: In addition: (j): The type tests described in ENATS 35-2 10.1.2 (c); dielectric tests consisting of lightning impulse chopped on the tail (LIC) in accordance with Clauses 13 and 14 of IEC60076-3 shall be carried out on every unit, i.e. shall be classed as a routine test.

10.1.2 Type tests: For any unit to be connected to the CE Electric UK electrical network, the full range of type tests shall be performed on the first unit of that type and rating from the given production facility. This requirement may be waived if CE Electric UK has had sufficient experience of transformers from the factory and the factory can supply type test criteria from similar units supplied to other UK Distribution Network Operators or National Grid.

10.6 Measurement of harmonics: "with 33kV applied to the normal tap" shall be replaced by "with Rated Voltage applied to the normal tap".

10.7 Measurement of zero phase sequence impedances(s) on three phase transformers: Table 3 Zero phase sequence impedance: The following addition shall be made to Table 3.

TABLE 3
Zero Phase Sequence Impedance

Emergency Rated Power (MVA)	Zero Phase Sequence Impedance (ohms/phase)	LV Neutral Current (A)
30	5-8	800

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10.10 Maximum noise levels - The following addition shall be made to Table 4

TABLE 4
Maximum Noise Levels

Rated Emergency Power (MVA)	ONAN Rating Sound Power Level (dBA)	CER Rating Sound Power Level (dBA)
30	67	84

10.10 Determination of sound levels:

A type test measurement for the CER rating shall be made with cooler banks fitted.

A type test measurement for the ONAN measurement shall be made with the cooler banks removed.

Routine ONAN measurements on all units shall be made without the cooler banks fitted and benchmarked against the ONAN type test measurement.

10.13 Site Tests: Site tests (i) through to (viii) inclusive shall be performed on every unit.

The following site tests shall also be performed:

(ix) FRA on 66kV units as described in 10.1.1 above

Vector group and ratio check

Check on pressure relief valve switch operation

12.2 Cooling medium: Where transformers contain insulating fluid then this fluid shall comply fully with the current CE Electric UK Technical Specification NPS/003/019: Technical Specification for Electrical Insulating Fluids for use in Plant & Switchgear. This requirement also applies to any fluid used for first fill and for the flushing of the unit

12.3 Cooler banks shall be either tank mounted or free standing as specified in Appendix 1: Schedule of Requirements.

12.4.1 Limiting Dimensions: The limiting dimensions of the 30MVA shall be the same as for the 20 and 40MVA option as shown on Figure 2.

12.4.2 Anti-vibration mountings: Anti-vibration mountings shall be matched and tuned to the individual transformer and cooler etc. The anti-vibration mountings shall be selected to match the longevity of the item they are supporting taking into account likely environmental conditions such as exposure to oil, sunlight etc. Details of the decisions and calculations supporting the anti-vibration measures shall be supplied to CE Electric UK.

13.1 Tanks and covers: The mechanical pressure tests shall be completed before any electrical tests.

13.1 Tanks and covers: The centre line shall be marked at base level on all sides, to facilitate alignment during handling, transport and location on site.

13.1 Tanks and covers: Working at Height Regulations: The requirements in ENATS 35-2 shall apply. Details of the provisions incorporated into the design to comply with the Working at Height Regulations must be agreed with CE Electric UK. Where work at height is unavoidable and access is to be provided by portable ladder, restraining brackets shall be incorporated into the transformer steelwork to locate and secure the ladder both at the top and the bottom. This is likely to include access to the Buchholz relay, winding temperature indicator thermocouples and connections, and CT terminations.

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13.1 Ladders used by CE Electric UK conform to BS 1129 and BS EN131 and are manufactured from either wood or glass-fibre. The dimensions of ladders used by CE Electric UK are listed below.

Rung diameter (mm)	22 (min) 80 (max)
Internal width of section: The minimum distance between sides will not be less than	235mm.

13.2 Surface Finish: The surface finish colour shall be uniform across the transformer and its auxiliaries and shall be a medium neutral grey.

13.3 Terminals: Where cable connections are required then separable plug-in cable terminations are preferred. Terminations should comply with CE Electric UK technical specifications:
NPS/002/015 Technical Specification for 11kV and 20kV Joints and Terminations
NPS/002/016 Technical Specification for 33kV Joints and Terminations
NPS/002/017 Technical Specification for 66kV and 132kV Cable Accessories

13.3 Terminals: Where outer cone separable connectors are used these shall be housed in an earthed metal enclosure that:

- (i) When installed on the switchgear provides physical protection, tested to 5J impact at the most vulnerable point on all exposed sides and at the vulnerable point on each exposed edge.
- (ii) Provides a minimum IP rating of IP21B in accordance with IEC 60529. For the purposes of this test the unscreened insulation is to be classed as a hazardous part that requires 20mm clearance from the end of the IP21B test probe.

13.3. Terminals: Gland plates shall be split across each row of gland holes to ease assembly.

13.3.1 HV terminals: The transformer shall NOT be fitted with Option (i) a three pole single gland compound filled box.

13.3.1 HV terminals: The facings on the transformer need not be arranged to allow for changing from outdoor bushings to cable boxes or vice-versa without any changes to component parts.

13.3.1 HV terminals: Table 5 Outdoor bushing details - The following additions shall apply:

TABLE 5:
Outdoor Bushing Details

Voltage (kV)	Live Metal to Earth Flashover distance (mm)	Air end Creepage Distance (mm)	Stem Diameter (mm)	Stem Length (mm)
20	500	550	20	70
66	700	1820	30	125

13.3.1 Outdoor bushings: All bushings shall be fitted with co-ordinating gaps set as specified in Table 9 of ENATS 35 – 3.

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13.3.1 HV terminals: Table 6 Fully Air Insulated Cable Box Minimum Clearances: The following additions shall apply:

TABLE 6
Fully Air Insulated Cable Box Minimum Clearances

Voltage (kV)	Live Metal to Earth Flashover distance (mm)	Air end Creepage Distance (mm)	Phase to Phase Clearance (mm)
20	220	550	430

13.3.1 HV terminals: Where cable connections are provided then a suitable supporting bracket equipped with cable cleats shall be provided for the type and number of cable specified in the Appendix 1.

13.3.1 HV terminals: Where separable connectors are used, cable disconnecting links shall be provided in a segregated oil filled chamber as described in ENATS 35-2 clause 13.3.

13.3.2: HV neutral: This clause in ENATS 35-2 shall also apply to 66kV star point neutrals.

13.3.2: HV neutral: Further to the requirements of ENATS 35-2 the HV neutral point shall be provided with an accessible isolatable link to allow testing. The HV neutral link should be enclosed in a lidded, dry metal box. A drawing of the proposed arrangement should be provided for approval. When option (iii) is selected the NDVT shall comply with the technical specification in Appendix 5 of this document.

13.3.3 LV terminations: Cable disconnecting links shall be provided in a separate oil filled chamber such that the links can be removed without draining any oil from the main oil system.

13.3.3 LV terminals: Where outer cone cable connectors are used they shall not be stacked more than two per bushing. There shall be the same number of connectors on each bushing.

13.3.3 LV terminations: Where cable connections are provided then a suitable supporting bracket for equipped with suitable cable cleats shall be provided to suit the type and number of cables specified in Appendix 1: Schedule of Requirements.

13.3.4 LV neutral: Further to the requirements of ENATS 35-2, an accessible isolatable link shall be provided on the neutral point of the LV winding to allow testing. The isolatable link arrangement shall allow the neutral point to be connected either solidly to earth or via a cable to a neutral earthing resistor. The LV neutral link arrangement shall be enclosed in a metal cable box preferably the same one used for the LV cable terminations. A drawing of the proposed arrangement should be provided for approval.

13.3.5 Unit auxiliary transformer: The tank mounted auxiliary transformer shall comply with Appendix 6 of this specification and all relevant parts of IEC 60076 and shall be connected as described in ENATS 35-2. Where the main transformer has a secondary voltage of 20kV the mounting flange for the auxiliary transformer shall have a facing of type J (Fig 24 of BS2562).

13.3.5 Unit auxiliary transformer: The option of a tank-mounted, cable connected auxiliary transformer will be considered by CE Electric UK. Any agreed option will need to satisfy CE Electric UK protection policy requirements and accessibility requirements.

13.4 On Load Tapchanger: The term; 'Raise' shall mean raise tap number and raise LV volts; The term 'Lower' shall mean lower tap number and lower LV volts.

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13.4.1 On Load Tapchanger: Operating mechanism: It shall be possible to use the tapchanger manual-operating handle easily from ground level.

13.4.2 On Load Tapchanger: Segregation of compartments: The oil actuated relay shall also operate in the event of loss of oil from the system.

13.4.2 On Load Tapchanger: Segregation of compartments: All breather pipework shall comply with the connection requirements of clause 8.2 of this specification.

13.5 Clearances to exposed conductors: (i) From fixed access points or ground level to live metal shall be 3100mm for voltages exceeding 33kV but not exceeding 66kV

13.5 Clearances to exposed conductors: The following additions shall be made to Table 7

TABLE 7
External Air Clearance

HIGHEST VOLTAGE FOR EQUIPMENT	20kV mm	66kV mm
Between live metal including neutral, and earth.	500	700
Between live metal of different phases and phase and neutral	430	780
From live metal to any oil pipe work including conservator	500	700

13.6.2 Cooling plant: Contactors controlling the power supply to the fans and/or pumps, which are operated from the winding temperature indicator panel in the local substation control room, shall be 110V AC. This supply shall be provided from a 240/55-0-55V transformer in the tap changer.

13.6.2 Cooling plant: If the transformer cooling system does not have an oil pump and there are four or more cooling fans, then in order to maintain the required level of flexibility, half of the cooling fans shall be controlled by what would have been the oil pump control circuit.

13.6.2 Cooling plant: Temperature related variable speed fan motors are preferred to avoid any step increases in noise-level.

13.6.3 Gas-and-oil actuated relay: All valves and relay test facilities shall be padlockable using CE Electric UK standard padlocks.

13.6.3 Gas-and-oil actuated relay: The relay shall also operate in the event of loss of oil from the system.

13.6.5 Winding temperature indicator: Primary transformer: A winding temperature indicator shall be provided to monitor the temperature of the LV winding. Where an additional winding temperature indicator is required this shall be as specified in Appendix 1: Schedule of Requirements.

13.6.5 Winding Temperature Indicators: Appendix 1 will specify whether a capillary winding temperature indicator or a thermocouple winding temperature sensor is required.

13.6.5 Winding temperature indicators; capillary type:

When capillary type winding temperature indicators are specified the remote indicators shall each have four sets of independently adjustable contacts to give separate control of fan(s) and pump(s) in addition to alarm and trip facilities.

The remote winding temperature indicators shall each provide an analogue facility comprising of a 0 to 10mA DC isolated analogue output for input into the CE Electric UK SCADA system. This output shall

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be capable of supplying a 1500 ohms load. 0mA shall correspond to 30°C and 10mA shall correspond to 150°C. This facility shall be derived from a linear variable differential transformer mechanically coupled to the winding temperature indicator movement.

13.6.5 Winding temperature indicators; thermocouple sensor type:

When a thermocouple winding temperature sensor is specified, a temperature measuring thermometer (PT100) shall be located in the top of the transformer. The output from the thermometer shall be terminated in weatherproof compartment using connectors capable of accepting a screened multi-core cable from the winding temperature indicator panel located in the substation control room.

13.6.5 Winding temperature indicators: thermocouple sensor type: The thermocouple shall be wired as shown in CE Electric UK drawing Y030S6701.

13.6.6 Current Transformers: Winding Temperature Indicators: Where a capillary type winding temperature indicator is specified a current transformer shall be provided on the centre termination of the LV winding to supply the thermometer pocket heater.

13.6.6 Current Transformers: Winding Temperature Indicators: Where a thermocouple sensor type is specified a current transformer shall be installed on a termination of the LV winding as specified in Appendix 1, and should be connected as shown in drawing Y030S6701. The current transformer on the LV winding shall supply both the winding temperature indicator and the automatic voltage control schemes. The current transformers shall comply with the following specification.

TX Size (mVA)	15	24	30	40	50
LV Winding Temperature CT Ratio (also used for voltage control)	750/1	1200/1	1500/1	2000/1	2500/1

13.6.6 Current Transformers: Automatic voltage control: Where capillary type winding temperature indicators are specified a separate current transformer shall be provided on one terminal of the transformer LV winding to supply the substation automatic voltage control scheme. The secondary connections of this current transformer should be terminated in the transformer marshalling kiosk.

13.6.6 Current transformers: Where specified in the schedule of requirements CTs shall be provided for protection schemes. These shall be accommodated within the bushing turret or cable box as appropriate.

13.6.6 Current transformers: Where winding temperature sensors are specified then a minimum of two wedding ring CTs (15/0.015A and 0.05VA) shall be provided for fan and pump current monitoring. These shall be connected in accordance with CE Electric UK drawing Y030S6701.

13.6.6 Current transformers: Where CTs are mounted in bushing turrets CT test loops shall be provided generally in accordance with Figure 2 of ENATS 35-3.

13.6.6 Current transformers: The items specified in clauses 13.6.6 shall allow the connection arrangements set out in CE Electric UK drawing Y030S6701.

13.6.7 Other Fittings: All breather pipework shall comply with the connection requirements of clause 8.2 above

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14.1 Drawings: In addition to the drawing items (i) to (ix), a drawing of the oil flow and valve locations and valve functions is required.

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4. References

4.1 External Documentation

Reference	Title
ENATS 35-2 Issue 5	Emergency Rated System Transformers 33/11.5kV Delta/Star and Star/Star Connected
ENATS 35-3 Issue 1	Continuous Maximum Rated (CMR) System Transformers (for use on systems up to 132kV)
IEC 60076-3	Power Transformers: Part 3: Insulation levels dielectric tests and external clearance in air
IEC 60529	Degree of protection provided by enclosures (IP Code)
BS 1129	Specification for portable timber ladders, steps, trestles and lightweight staging.
BS EN 131-2	Ladders – Part 2: Specification for requirements, testing and marking.
BS2562	Cable boxes for transformers and reactors

4.2 Internal documentation

Reference	Title
Y030S6701	Supply point TX FDR/TX Transformer control Ashridge 852 Plus-P (panel mounted) HV and LV winding temperature indicators.
NPS/002/015	Technical Specification for 11kV and 20kV Joints and Terminations
NPS/002/016	Technical Specification for 33kV Joints and Terminations
NPS/002/017	Technical Specification for 66kV and 132kV Cable Accessories
NPS/003/019	Specification for Electrical Insulating Fluids for use in CE Electric UK Plant & Equipment.

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4.3 Amendments from Previous Version

Section	Subject	Amendments
2.	Scope	References to NEDL and YEDL replaced by CE Electric UK
2.1	Contents	Section added
3..2	Technical Specification	4.2 Graphs of CMR and ONAN versus ambient required at tender stage 5.6 Confirmation of method of capitalisation of load losses 6. Neutrals added to vector groups 6. Requirements for changeover links on 33kv TX removed 7.1 Rating Plate: Additional information required stated 8.2 Exclusion of refrigerator type breather removed 10.1 Requirement for 1/3 octave sound frequency test removed 10.1.2 Type tests. Conditions stated for waiving type-test requirement 13.1 Requirements with regard to Working at Height Regulations stated 13.2 Surface Finish: Colour added. 13.3 Requirement to bring neutral point out to disconnecting link added 13.3.2 Option for HV NDVT added. Spec added to as Appendix 5 13.3.1 Clearances for unfilled cable box added 13.3 References to NPS specifications for cable terminations added 13.3 Requirements added with regard to outer cone cable connectors 13.3 Requirements for cable gland plates added 13.3.1 Outdoor bushing: Requirements for co-ordinating gaps added 13.3.5 Tank mounted auxiliary transformer: Requirement clarified. 13.4 Requirement of weir around in-tank tap changer removed 13.6.2 Cooling plant: Option of variable speed motors added 13.6.3 Gas-and-oil relay: Requirement for loss of oil protection 13.6.5 WTI: Requirements for Capillary / thermocouple types clarified 13.6.6 CTs: Requirements for WTI and AVC stated 13.6.6 CTs: Requirements for test loops stated
4.3	Amendments from Previous Version	Section added

5. Definitions

Term	Definition
NONE	

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
K Midgley	Standards Engineer	K Midgley	23/11/2009

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	Head of System Strategy	M Nicholson	24/11/2009

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
D Blackledge	Plant and Switchgear Standards Manager	D Blackledge	23/11/2009

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	30/11/2009

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	25/11/2009

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	25/11/2009

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Appendix 1 - Schedule Of Requirements

Clause	Description	
1.2.2	Ambient temperature for Emergency Rating °C	30
4.1	Rated power at Emergency Rating MVA HV Nominal Voltage kV LV Nominal Voltage kV	* 15 / 24 / 30 / 40 / 50
5.4	Tapping range a) (+/-8x1.25%) +/-10% in 16 steps or b) (+4x1.43% / -12x1.43%) +5.72 / -17.16% in 16 steps	a / b
5.5	Impedance envelope for 12/24MVA Impedance envelope for 15/30MVA Or CEEUK specific requirements	1 (b) / 1 (c) 1 (b) / 1 (c)
5.6	Loss Capitalization Values (i) No-load Loss £/kW (ii) Load Loss at 50% CER rating £/kW Lifetime Cost = Purchase price + ((No load loss kW x No load £/kW) + (Load loss kW x Load loss £/kW)). Note that the CE Electric UK loss £/kW figures incorporate utilisation factor and timespan	
6	Vector Group (Connection Symbol): Include phase displacement numbers and neutral symbols where appropriate.	
6	Vector Links required If yes, alternative vector group	* Yes / No
8.1	Neutral conductor and terminal intended to carry load	* Yes / No
8.2	Oil preservation system If free breathing, breather type	* Free breathing / Diaphragm conservator * Desiccant / refrigerating
8.3	Generator transformer load rejection per 8.3	* Yes / No
10.1.1	FRA test required at works and on site	* Yes / No
10.1.3	Short circuit test required Third harmonic test required	* Yes / No * Yes / No
12.3	Cooler Banks: a) Separate b) Tank mounted	Separate / Tank Mounted

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Clause	Description	
12.4.1	Separate cooler bank required Height limitationmm Valve & blanking plates required at alternative oil inlet	* Yes / No * 4880 / 4270 * Yes / No
12.4.2	Anti-vibration mountings required	* Yes / No
12.5	Auxiliary supply voltage	* 400 V 3 ph / 230 V 1 ph
13.3	Position of cable boxes	* Same / opposite side of tank Manufacturer to decide
13.3.1	HV terminal arrangement NVD capacitor taps required Capacitance required pF Surge arresters required Brackets for surge arresters required Number and size of single core cables to be terminated	* (i) / (ii) / (iii) / (iv) / (v) * Yes / No * 60 / 90 / 140 * Yes / No * Yes / No
13.3.2	HV neutral terminal arrangement Size of single core cable to be terminated	* (i) / (ii) / (iii) / (iv)
13.3.3	LV Terminal arrangement Number and size of single core cables to be terminated.	* Cable box / separable connectors / open bushings
13.3.4	LV neutral terminal arrangement Size of single core cable to be terminated LVN current transformer(s) required Details of LVN current transformer(s)	* (i) / (ii) / (iii) / (iv) * Yes / No
13.3.5	Additional flange required for auxiliary transformer	* Yes / No
13.4.1	Max / min tap position indicator required	* Yes / No
13.6.5	Winding temperature indicators (WTI) Capillary or thermocouple sensors type. WTI on HV winding WTI on LV winding	 Yes/No Yes/No
13.6.6	Winding temperature indicator CT position HV LV Tertiary Additional current transformer accommodation required: CTs for voltage control Details: CTs for protection schemes Details:	 * Yes / No

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Appendix 2 - Self-Certification Conformance Declaration For Emergency Rated System Transformers: 33/11.5kv

Appendix 2: Self-Certification Conformance Declaration

CLAUSE BY CLAUSE CONFORMANCE WITH ENA TS 35-2

Transformers covered by ENA TS 35-2 shall comply with the latest issues of the relevant International and British Standards. ENA TS 35-2 is intended to amplify and/or clarify the requirements of those standards.

This check sheet identifies the clauses in ENA TS 35-2 and the clauses of IEC 60076-1. The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes N/A = Clause is not applicable / appropriate to the product Cs1 = The product conforms fully with the requirements of this clause Cs2 = The product conforms partially with the requirements of this clause Cs3 = The product does not conform to the requirements of this clause	Instructions for completion - When Cs2 or Cs3 is entered, the reason for non-conformance shall be entered - Where options are specified, the remark should identify the option offered by the manufacturer - The manufacturer should complete the schedule items in the Remarks column. Additional sheets may be referred to where more detail is required.
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Clause / Sub-clause	Requirement	Conformance Code IEC 60076-1	Conformance Code ENA TS 35-2	Schedule item	Remarks
1	Scope and service conditions			N/A	
1.1	Scope				
1.2	Normal Service Conditions		N/A	N/A	
1.2.2	Provision for Unusual Service Conditions			Value of cooling air temp °C	
2	Normative references			N/A	
3	Definitions		N/A	N/A	
3.1	General				
3.2	Terminals and Neutral Point		N/A	N/A	
3.3	Windings		N/A	N/A	
3.4	Rating		N/A	N/A	

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Clause / Sub-clause	Requirement	Conformance Code IEC 60076-1	Conformance Code ENA TS 35-2	Schedule item	Remarks
3.5	Tappings		N/A	N/A	
3.6	Losses and No-load Current		N/A	N/A	
3.7	Short-circuit Impedance and Voltage Drop		N/A	N/A	
3.8	Temperature Rise		N/A	N/A	
3.9	Insulation		N/A	N/A	
3.10	Connections		N/A	N/A	
3.11	Kinds of Tests		N/A	N/A	
3.12	Meteorological Data with Respect to Cooling		N/A	N/A	
4.1	Rating			Emergency rated power (MVA)	
4.2	Loading cycle			Graph of CER vs. ambient CMR at rated temp (MVA)	
4.3	Preferred values of rated power				
5	Tapped windings		N/A	N/A	
5.1	Notation of tapping range				
5.2	Tapping voltage, current, etc		N/A	N/A	
5.3	Tapping power		N/A	N/A	
5.4	Specification of tappings			Tapping range offered	
5.5	Specification of short-circuit impedance			Graph of impedance vs. tap position	
5.6	Load loss and temperature rise			No load loss kW Guaranteed load loss at 50% of CER rating kW	
6	Vector Group (connection and phase displacement symbol)			Vector group (connection symbol) Alternative vector group where links are provided.	
7	Rating plates		N/A	Additional items included on rating plate?	
7.1	Information to be given in all cases				
7.2	Additional information to be given		N/A	N/A	
8	Miscellaneous requirements			Is neutral connection dimensioned for load current?	
8.1	Dimensioning of neutral connection				

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Clause / Sub-clause	Requirement	Conformance Code IEC 60076-1	Conformance Code ENA TS 35-2	Schedule item	Remarks
8.2	Oil preservation system			Description of oil preservation system.	
8.3	Load rejection on generator transformers			Is the transformer suitable for generator transformer load rejection?	
9	Tolerances		N/A	N/A	
10 10.1	Tests General requirements			Dielectric test level Rated lightning impulse (kV peak) 50Hz withstand voltage (kV rms)	
10.1.1	Routine tests			Guaranteed sound power level at ONAN dB(A) Frequency response analysis? Lightning impulse chopped tail?	
10.1.2	Type tests			Are type test results from an identical unit available? Guaranteed sound power level at CER dB(A)	
10.1.3	Special tests			Guaranteed and expected change of impedance after short-circuit test Special tests included in offer	
10.2	Measurement of winding resistance		N/A	N/A	
10.3	Measurement of winding ratio and phase displacement		N/A	N/A	
10.4	Measurement of short-circuit impedance and load-loss			N/A	
10.5	Measurement of no-load loss and current		N/A	Guaranteed no-load loss kW	
10.6	Measurement of harmonics		N/A	N/A	
10.7	Measurement of ZPSI			Guaranteed ZPSI Ω/ph	
10.8	Tests on OLTC		N/A	N/A	
10.9	Temperature Rise at Emergency Rating	N/A		N/A	

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Clause / Sub-clause	Requirement	Conformance Code IEC 60076-1	Conformance Code ENA TS 35-2	Schedule item	Remarks
10.10	Determination of Sound Levels	N/A		N/A	
10.11	Pressure test	N/A		N/A	
10.12	Frequency Response Analysis	N/A		Is frequency response analysis is included in offer?	
10.13	Site tests	N/A		N/A	
11	EMC		N/A	N/A	
12	Transformer details	N/A		N/A	
12.1	No. of phases and frequency				
12.2	Cooling medium	N/A		Cooling medium offered.	
12.3	Cooling	N/A		Cooling type at CER. Tank mounted or free standing coolers.	
12.4	Limiting dimensions	N/A		Does the transformer meet requirements of 12.4.1?	Yes/ No
12.4.1	Separate cooler bank arrangement				
12.4.2	Anti-vibration mountings	N/A		Percentage isolation value and description of AV mountings	
12.5	Auxiliary supply voltage	N/A		Supply voltage for tap-changer and cooler	

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Clause / Sub-clause	Requirement	Conformance Code IEC 60076-1	Conformance Code ENA TS 35-2	Schedule item	Remarks
13 13.1	Construction details Tanks and covers	N/A		Is offer for bolted or welded cover? Total oil required (including cooling system). litres Volume of oil to be removed: (i) to gain access to core earthing link (ii) to effect in-situ change of HV bushing Filling medium for transport Total weight as installed in service including cooler plant, all fittings and oil. Tonnes	
13.1.1	Gaskets	N/A		Description of gasket material and thickness	
13.2	Surface finish	N/A		Finish colour offered	
13.3	Terminals	N/A		Position of HV and LV terminals	
13.3.1	HV terminals	N/A		Type of terminals offered?	
13.3.2	HV neutral terminal	N/A		Type of terminal offered?	
13.3.3	LV termination	N/A		Type of terminals offered?	
13.3.4	LV neutral termination	N/A		Type of terminal offered?	

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Clause / Sub-clause	Requirement	Conformance Code IEC 60076-1	Conformance Code ENA TS 35-2	Schedule item	Remarks
13.3.5	Unit auxiliary transformer (UAT)	N/A		Is flange and valve for UAT included	
13.4	On-load tap-changer	N/A		N/A	
13.4.1	Operating mechanism	N/A		Description of motor protection Is a max/min tap position indicator included?	
13.4.2	Segregation of compartments	N/A		N/A	
13.4.3	Method of operation	N/A		N/A	
13.5	Clearance to exposed conductors	N/A		N/A	
13.6	Fittings	N/A		N/A	
13.6.1	Conservator	N/A		N/A	
13.6.2	Cooling plant	N/A		Cooling loss (kW) Number of pumps Continuous rating of pump motor shaft kW Starting current of pump motor(A) Contactor control voltage (V) Number of fans Continuous rating of fan motor shaft kW Starting current of fan motor(A) Contactor control voltage (V) Fan diameter mm Speed of fans rpm	

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Clause / Sub-clause	Requirement	Conformance Code IEC 60076-1	Conformance Code ENA TS 35-2	Schedule item	Remarks
13.6.3	Gas-and-oil actuated relay	N/A		N/A	N/A
13.6.4	Pressure relief device	N/A		N/A	
13.6.5	Winding temperature indicators	N/A		Description of provision of Winding temperature indicator:	
13.6.6	Current transformers	N/A		N/A	
13.6.7	Other fittings	N/A		N/A	
13.6.8	Valves	N/A		N/A	
13.7	Marshalling/control box	N/A		Means of condensation protection Description of security	
13.8	Interconnecting cables	N/A		N/A	
14	Documentation	N/A		Proposed maintenance schedule	
14.1	Drawings	N/A		Electronic drawing format	
14.2	Assembly, operation and maintenance instructions			Electronic format for instruction manual	

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Appendix 3 - Manufacturers And Places Of Manufacture, Testing And Inspection

Item	Manufacturer's Drawing Number and/or Type Designation	Manufacturer	Place of Manufacture	Place of Testing and Inspection
Transformers complete On-load tap-change equipment HV Bushings LV Bushings Neutral Bushings Radiators Pipework Expansion Devices Oil valves Oil Pumps Oil Pump Motors Fans Fan Motors Dehydrating Breather Gas and Oil Actuated Relay(s) Outdoor Marshalling/Control Box Temperature Indicating Devices Auxiliary Transformer Material for Anti-Vibration Mounting				
Any deviation from this Schedule shall be notified in writing as soon as possible for the Purchaser's approval.				

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Appendix 4 – Self Certification Conformance Declaration

CLAUSE BY CLAUSE CONFORMANCE WITH CEEUK Specification NPS/003/012

Clause / Sub-clause	Requirement	Conformance Code NPS/003/012	Schedule item	Remarks
(3.2.2) 4.1	Rating		Rating ambient temp (30°C)	
(3.2.2) 4.1	Rating		15/30 MVA rating available	
(3.2.2) 4.2	Loading Cycle		Additional graphs	
(3.2.2) 5	Expression of tapping range		Nomenclature change	
(3.2.2) 5.5	Short circuit impedance		Impedance Figure 1 added data.	
(3.2.2) 5.6	Loss capitalisation basis. As described in Appendix 2 of this document.		Capitalisation formula	
(3.2.2) 6	Connection and phase displacement symbols		Vector groups requiring vector changeover links	
(3.2.2) 7.1	Additional rating plate data required		Add data to rating plate	
(3.2.2) 8.2	Oil preservation system		Pipe thread 1" BSP	
(3.2.2) 10.1	Tests: General		Mechanical tests before electrical tests.	
(3.2.2) 10.1	Tests General		Additional values in Table 2 11kV tested at 95kV/38kV Addition of 20kV and 66kV criteria	
(3.2.2) 10.1.1	Routine tests		2 x Frequency Response Analysis tests required for 66kV units	
(3.2.2) 10.1.1	Routine tests		Added routine test: Lightning impulse chopped wave test	

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Clause / Sub-clause	Requirement	Conformance Code NPS/003/012	Schedule item	Remarks
(3.2.2) 10.1.2	Type tests		Full type tests on first unit of each type from each production facility.	
(3.2.2) 10.6	Measurement of Harmonics		"33kV" replaced with "rated voltage"	
(3.2.2) 10.7	Zero phase sequence currents		Addition of zero phase sequence currents for 30MVA rating	
(3.2.2) 10.10	Maximum noise levels		Addition of maximum noise levels for 30MVA rating	
(3.2.2) 10.10	Determination of sound levels		CER type test with radiators fitted and all cooling on Benchmark ONAN type test without radiators fitted Routine ONAN noise measurement without radiators fitted	
(3.2.2) 10.13	Site tests		Additional site tests: FRA, Vector group, ratio, pressure relief switch	
(3.2.2) 12.2	Cooling medium		Cooling medium to NPS/003/019	
(3.2.2) 12.3	Cooling		Cooler banks: Free standing or tank mounted options available.	
(3.2.2) 12.4.1	Separate cooler bank arrangement.		Limiting dimensions of 30MVA rated unit added	
(3.2.2) 12.4.2	Anti vibration mountings requirements		Additional requirement Calculations to be supplied.	
(3.2.2) 13.1	Tanks and covers		Mechanical tests to precede electrical tests	

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Clause / Sub-clause	Requirement	Conformance Code NPS/003/012	Schedule item	Remarks
(3.2.2) 13.1	Tanks and covers		Centre line at ground level and on all sides.	
(3.2.2) 13.1	Tanks and covers		Working at height arrangements; evidence of consideration, options and conclusion	
(3.2.2) 13.1	Tanks and covers		Ladder securing arrangements for short time inspection/work	
(3.2.2) 13.2	Surface finish		Medium neutral grey	
(3.2.2) 13.3	Terminals		Compliance with CE Electric UK cable termination specifications	
(3.2.2) 13.3	Terminals		Earthed metal enclosure for outer cone connectors	
(3.2.2) 13.3	Terminals		Split gland plates	
(3.2.2) 13.3.1	HV Terminals		Exclusion of compound filled box option	
(3.2.2) 13.3.1	HV Terminals		No requirement for convertibility between external bushing and cable box	
(3.2.2) 13.3.1	HV Terminals		Additions to Table 5 – outdoor bushing details	
(3.2.2) 13.3.1	HV Terminals		External bushing coordinating gap	
(3.2.2) 13.3.1	HV Terminals		Additional air insulated cable box clearance.	
(3.2.2) 13.3.1	HV Terminals		Provision of cable cleats on supporting bracket	
(3.2.2) 13.3.1	HV Terminals		Cable disconnecting links where inner cone connectors are used	

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Clause / Sub-clause	Requirement	Conformance Code NPS/003/012	Schedule item	Remarks
(3.2.2) 13.3.2	HV Neutral		66kV star point neutral	
(3.2.2) 13.3.2	HV Neutral		Isolatable links	
(3.2.2) 13.3.3	LV Terminations		Cable disconnecting links	
(3.2.2) 13.3.3	LV Terminations		Stacking outer cone connections	
(3.2.2) 13.3.3	LV Terminations		Provision of cable cleats on supporting bracket	
(3.2.2) 13.3.4	LV Neutral		Isolatable links	
(3.2.2) 13.3.5	Unit auxiliary transformer		Provision of tank mounted auxiliary transformer	
(3.2.2) 13.3.5	Unit auxiliary transformer		Cable connection option of auxiliary transformer	
(3.2.2) 13.4	On load tap changer		'Raise' and 'Lower' tap change terminology	
(3.2.2) 13.4.1	On load tap changer		Tap change operating handle ergonomically accessible at ground	
(3.2.2) 13.4.2	On load tap changer		Relay operation for loss of oil	
(3.2.2) 13.4.2	On load tap changer		Breather pipework termination	
(3.2.2) 13.5	Clearance to exposed conductors		Clearance for voltages above 33kV not exceeding 66kV	
(3.2.2) 13.5	Clearance to exposed conductors		20kV & 66kV clearances added to Table 7	
(3.2.2) 13.6.2	Cooling plant		Control supply 110V from 240/55-0-55V transformer	
(3.2.2) 13.6.2	Cooling plant		Staged operation of fans where no oil pumps	
(3.2.2) 13.6.2	Cooling plant		Variable speed fan motors	
(3.2.2) 13.6.3	Gas and oil actuated relay		Sampling and testing points lockable	

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Clause / Sub-clause	Requirement	Conformance Code NPS/003/012	Schedule item	Remarks
(3.2.2) 13.6.3	Gas and oil actuated relay		Relay operation for loss of oil	
(3.2.2) 13.6.5	Winding Temperature Indicators		winding temperature indicator or temperature sensor	
(3.2.2) 13.6.5	Winding Temperature Indicators		winding temperature indicator criteria	
(3.2.2) 13.6.5	Winding Temperature Indicators		Temperature sensor criteria	
(3.2.2) 13.6.5	Winding Temperature Indicators		Wired as per Y030S6701	
(3.2.2) 13.6.6	Current Transformers		CTs for voltage control scheme In bushing turret or cable box	
(3.2.2) 13.6.6	Current Transformers		CTs for protection schemes In bushing turret or cable box	
(3.2.2) 13.6.6	Current Transformers		Min of 2 wedding ring CTs connected as specified	
(3.2.2) 13.6.6	Current Transformers		Wired as per Y030S6701	
(3.2.2) 13.6.7	Other Fittings		Breather pipework termination	
(3.2.2) 14.1	Additional drawings		N/A	

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Appendix 5 – Specification for Transformer Tank Mounted Neutral Displacement Voltage Transformer (NDVTs)

A5-1 Overview

A5.1.1 ENATS 35-2 allows for flange mounting of an outdoor voltage transformer for neutral displacement protection. The NDVT primary winding shall be connected between the transformer winding star point and earth.

A5.1.2 The NDVT shall be designed, constructed and tested in accordance with BS EN 60044:1999 Instrument transformers – Part 2 Inductive voltage transformers

A5-2. Electrical Performance Requirements

A5.2.1 Highest voltage for equipment (Um) 36 kV

A5.2.2 Secondary winding output voltage 63.5 V

A5.2.3 Minimum rating 50 VA

A5.2.4 Accuracy Class 6P

A5.2.5 Basic Insulation Oil Immersed

A5-3. General Construction and Fittings

A5.3.1: The oil in the NDVT shall be separated from the oil in the main transformer by means of an internal barrier that shall be pressure tested in accordance with section 10.11 of ENATS 35-2, in both directions.

A5.3.2: The NDVT shall incorporate the following features:

Oil preservation system in accordance with Sub clause 8.2 of section 3.3.2 of NPS/003/012

Tank and covers in accordance with Sub clause 13.1 of section 3.3.2 of NPS/003/012 and within the constraints CE Electric UK drawing number Y213X99312.

Oil level indicator

Rating and diagram plate

Terminal box with removable, padlockable cover. This terminal box shall contain links as required to comply with the winding arrangements specified in section A5-4 below. These links shall be clearly identified.

A cable gland for a 2 core multicore cable

A5-4. Winding Arrangements

A5.4.1: The earthing of the primary winding shall be independent of, and segregated from, the earthing of the secondary winding.

A5.4.2: Primary winding terminal "N": The primary winding neutral point shall be provided with an accessible isolatable link to allow testing. The primary winding neutral link should be enclosed in a lidded, dry metal box. A drawing of the proposed arrangement should be provided for approval.

A5.4.3: The primary winding terminal "A" shall be connected to a HV bushing by a flexible lead.

This bushing shall connect the NDVT to the main transformer HV neutral and shall be within the main transformer oil system.

It shall be possible to easily disconnect this flexible lead to the bushing for the purpose of testing the main transformer.

A5.4.4: The secondary winding terminals "a" and "n" shall be connected to links (typically GEC RS20 type) or equivalent in the terminal box. The "n" terminal winding neutral point shall be provided with an accessible isolatable link to allow testing. The secondary winding neutral link should be enclosed in a

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lided, dry metal box. A drawing of the proposed arrangement should be provided for approval.

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A5-5. NDVT Conformance Declaration

The conformance declaration table below shall be completed for each unit.

Clause	Description	Confirmation/Comment/Description
A5.1.1	NDVT primary winding	
A5.1.2	Compliance with BS EN 60044:1999 Instrument transformers – Part 2 Inductive voltage transformers	
A5.2.1	Highest voltage for equipment (Um)	
A5.2.2	Secondary winding output voltage	
A5.2.3	Burden rating	
A5.2.4	Accuracy class	
A5.2.5	Basic insulation type	
A5.3.1	Barrier pressure test	
A5.3.2	Oil preservation system in accordance with section 8.2 of NPS-003-012	
	Tank and covers in accordance with section 13 of NPS-003-012 and within the constraints CEEUK drawing number Y213X99312	
	Oil level indicator	
	Rating and diagram plate	
	Terminal box with removable, padlockable cover.	
	This terminal box links.	
	Cable gland for a 2 core multicore cable	
A5.4.1	Independent, segregated primary and secondary earthing	
A5.4.2	Primary winding terminal "N"	
A5.4.3	Primary winding terminal "A"	
A5.4.4	Secondary winding terminals "a" and "n"	

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Appendix 6 – Specification for Transformer Tank Mounted Auxiliary Transformers

A6.1 Overview

ENATS 35-2 allows for flange mounting of an outdoor auxiliary transformer. The auxiliary transformer shall be designed, constructed and tested in accordance with ENATS 35-1 - Distribution transformers, except where varied by this specification.

A6.2. Electrical Performance Requirements

A6.2.1: Input voltage for equipment (Um) to match the main transformer secondary voltage.

A6.2.2: Output voltage 415V at no load.

A6.2.3: Vector Group Dyn11

A6.2.4: Rating 25kVA

A6.2.5: Tappings Not required

A6.2.6: Impedance Voltage (at 75°C) 4.5% of rated power

A6.3. Mounting Arrangements

A6.3.1: The flange facing and mounting arrangement need not comply with the flange facings specified in ENATS 35-2, or in ENATS 35-1, but shall optimise the interface between the main unit and the auxiliary transformer.

A6.3.2: Mounting Arrangements: The auxiliary transformer shall be rigidly supported by the flange mounting arrangement. Additional bracings shall be provided between the auxiliary and main transformer if necessary to ensure adequate and rigid support.

A6.4. LV Terminal Arrangements

A6.4.1: An LV Fuseboard shall be mounted on the auxiliary transformer and the LV terminals of the auxiliary transformer shall be connected to this LV fuseboard. The means of connection shall minimise any risk from failure of this connection(s).

A6.4.2: The LV fuseboard shall conform fully to Form 4 Type 2 requirements of BS EN 60439-1, Low-voltage switchgear and controlgear assemblies – Type-tested and partially type-tested assemblies. All terminals and exposed live metalwork shall be shrouded.

A6.4.3: The LV fuseboard shall be designed and constructed to be suitable for outdoor use. The fuseboard, its components (excluding insulating parts) and its mountings shall not require maintenance for a period of at least 30 years in a polluted / coastal environment according to EN ISO 12944-2 Category C4.

A6.4.4: The fuseboard housing shall have a padlockable access door/non-bolted cover; suitable for use with a padlock hasp of 3-6mm diameter.

A6.4.5: The auxiliary transformer LV phase connections shall be terminated, in the LV fuseboard, onto three 100A fuse units that shall be in accordance with Type 3 of BS 7657 (1993); each of which shall be fused with a 60A fuse that shall be in accordance with Type II b of BS1361 (1971).

A6.4.7: The auxiliary transformer LV fuses shall be ergonomically accessible from ground level and be mounted not less than 900mm and not more than 1600mm above ground level.

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A6.4.8: The auxiliary transformer LV neutral connections shall be terminated onto an arrangement with two independently removable, bolted links inside the LV fuseboard:

- (a) the outgoing side of one link shall be connected to a designated earth connection point on the auxiliary transformer
- (b) the outgoing side of one link shall be connected to the outgoing LV cable

A6.4.9: The underside of the LV fuseboard housing shall have a suitably positioned 20mm diameter cable access hole, equipped with a compression gland for a 4-core armoured 25mm² cable that will be used to connect to the outgoing sides of the LV fuses.

A6.4.10: The LV fuseboard shall be equipped with all necessary connections/lugs to allow termination of a 4-core armoured 25mm² cable used to connect to the outgoing sides of the LV fuses

A6.5. Earthing Terminal Arrangement

A6.5.1: An earthing connection shall be provided on the exterior of the auxiliary transformer. This connection shall be designed to accept neutral/earth connection(s) from the bolted link in the LV fuseboard described above.

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A6-6. Auxiliary Transformer Conformance Declaration

The conformance declaration table below shall be completed for each unit.

Clause	Description	Confirmation/Comment/Description
A6.1.1	Conformance with ENATS 35-1	
A6.2.1	Input voltage	
A6.2.2	Output voltage	
A6.2.3	Vector group	
A6.2.4	Rating	
A6.2.5	Tappings	
A6.2.6	Impedance voltage (at 75°C)	
A6.3.1	Flange facing	
A6.3.2	Mounting arrangements	
A6.4.1	LV terminals & connection	
A6.4.2	Form 4 Type 2 requirements of BS EN 60439-1	
A6.4.3	Withstand 30 years in EN ISO 12944-2 Category C4	
A6.4.4	Padlockable door/cover	
A6.4.5	LV fuse & link arrangements	
A6.4.6	Neutral connection	
A6.4.7	Cable hole & gland	
A6.5.1	Earth terminal arrangement	